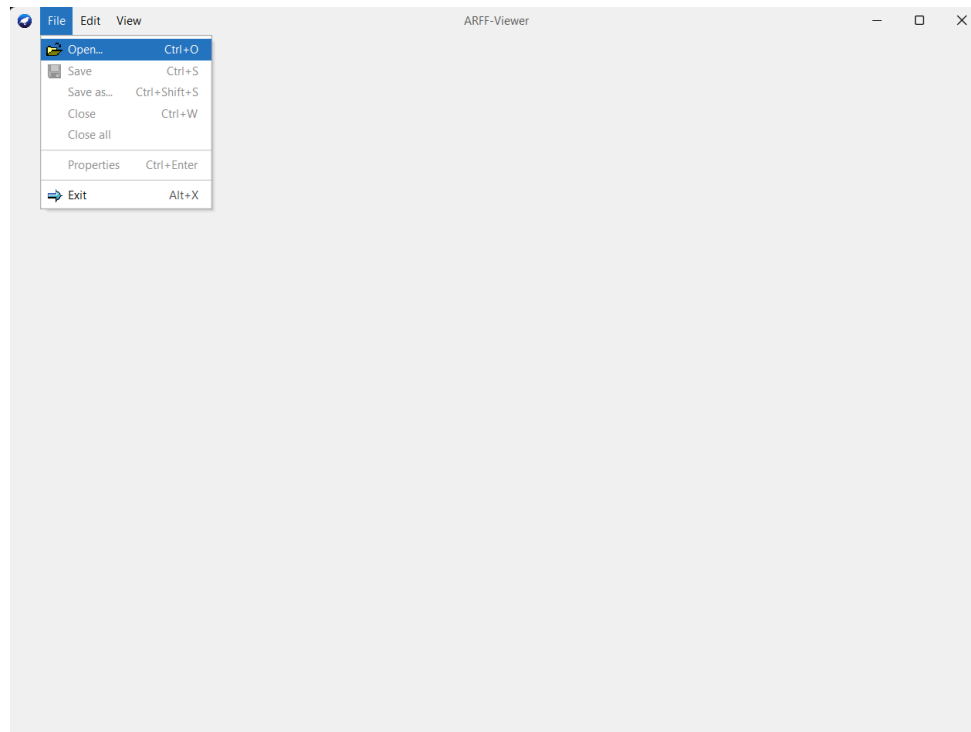
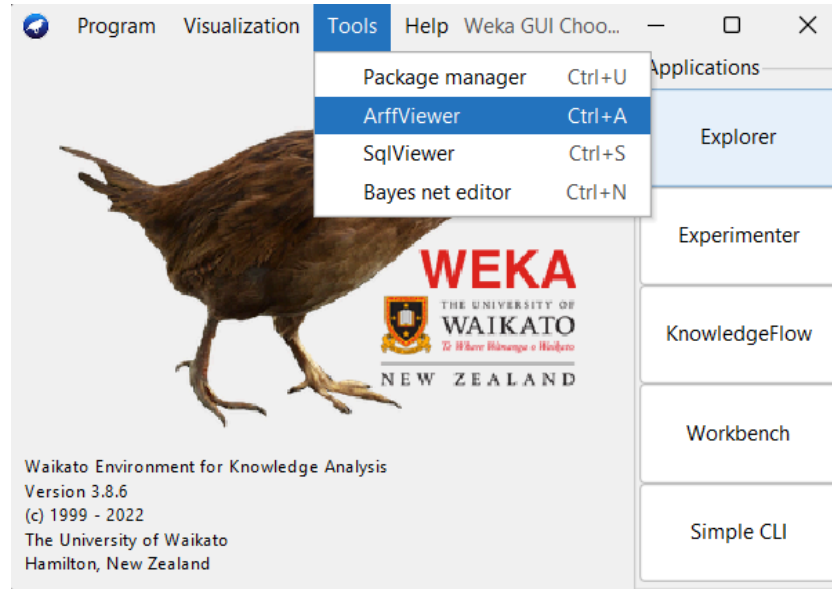
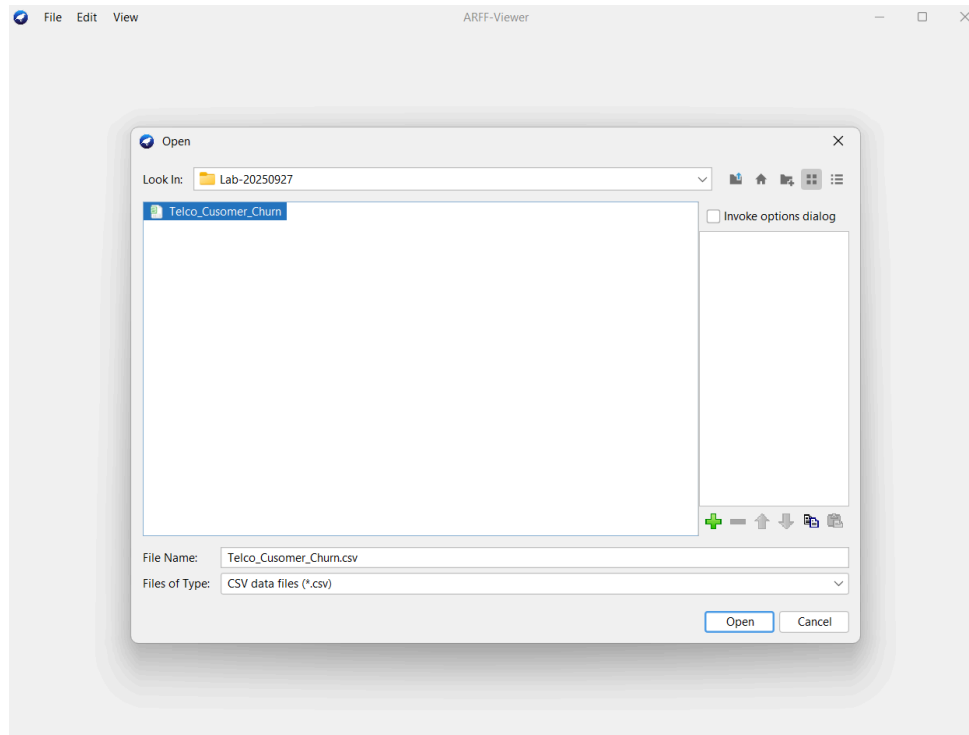


Big Data Analytics

Name: Reyes, Francis Albert M.	Date: Sep 13, 2025
Section: CS41S1	Prof. John Richard Kho

DATA UNDERSTANDING





ARFF-Viewer - C:\Users\rfra\OneDrive\Desktop\bda_cs41s1\Lab-20250927\Telco_Cusomer_Churn.csv

Telco_Cusomer_Churn.csv

Relation: Telco_Cusomer_Churn

No.	1: customerID Nominal	2: gender Nominal	3: SeniorCitizen Numeric	4: Partner Nominal	5: Dependents Nominal	6: tenure Numeric	7: PhoneService Nominal	8: MultipleLines Nominal	9: InternetService Nominal	10: OnlineSecurity Nominal	11: OnlineBackup Nominal	12: D
1	7590-VHVEG	Female	0.0	Yes	No	1.0	No	No phone servi...	DSL	No	Yes	No
2	5575-GNVDE	M										Yes
3	3668-QPYBK	M										No
4	7795-CFOCW	M										Yes
5	9237-HQITU	Fe										No
6	9305-CDSKC	Fe										Yes
7	1452-KIOVK	M										No
8	6713-OKOMC	Fe										No
9	7892-POOKP	Fe										Yes
10	6388-TABGU	M										No
11	9763-GRSKD	M										No
12	7469-LKBCI	M									internet servi...	No ir
13	8091-TTVAX	M										Yes
14	0280-XJGEX	M										Yes
15	5129-JLPIS	M										Yes
16	3655-SNQYZ	Fe										Yes
17	8191-XWSZG	Fe									internet servi...	No ir
18	9959-WOFKT	M										Yes
19	4190-MFLUW	Fe										Yes
20	4183-MYFRB	Fe										Yes
21	8779-QRDMV	M										Yes
22	1680-VDCWW	M									internet servi...	No ir
23	1066-JKSGK	M									internet servi...	No ir
24	3638-WEABW	Fe										No
25	6322-HRPFA	M										No
26	6865-JZKNO	Fe										No
27	6467-CHFZW	M										No
28	8665-UTDHz	Male	0.0	Yes	Yes	1.0	No	No phone servi...	DSL	No	Yes	No
29	5248-YGIJN	Male	0.0	Yes	No	72.0	Yes	Yes	DSL	Yes	Yes	Yes
30	8773-HHUOZ	Female	0.0	No	Yes	17.0	Yes	No	DSL	No	No	No
31	3841-NFEEX	Female	1.0	Yes	No	71.0	Yes	Yes	Fiber optic	Yes	Yes	Yes
32	4929-XIHWW	Male	1.0	Yes	No	2.0	Yes	No	Fiber optic	No	No	Yes

Save In: Lab-20250927

Invoke options dialog

File Name: Telco_Cusomer_Churn

Files of Type: Arff data files (*.arff)

Save Cancel

REFLECTION:

A. Compared to the original file format (.csv), what changes were made after converting the file to .arff? Kindly elaborate on your observation. (10 pts)

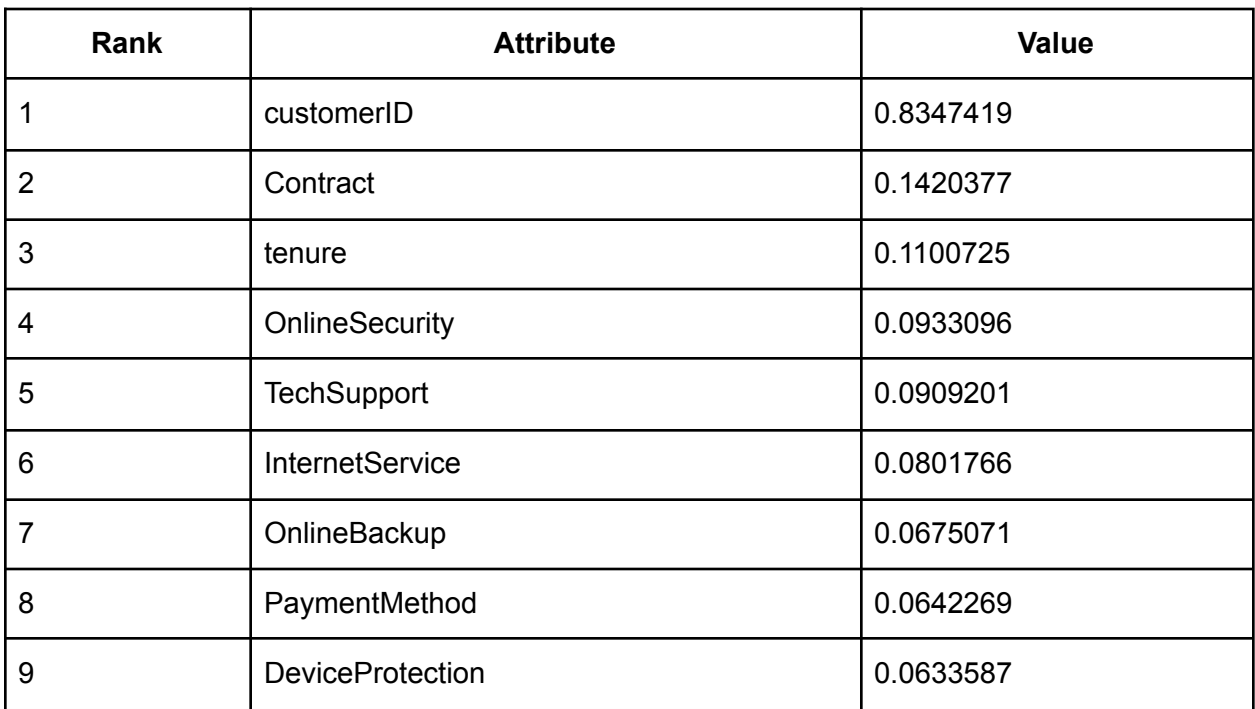
The conversion from csv to arff was easy because I just followed the steps. The csv only showed plain rows and columns of data while the arff file became more organized. It now separates the dataset description from the actual data and it also shows the type of each attribute whether it is numeric like tenure or monthly charges or nominal like gender contract or payment method. This makes it easier for Weka to process the file correctly.

DATA PREPARATION



The screenshot shows the 'Weka Explorer' window. The 'Preprocess' tab is selected. Below the menu bar are buttons for 'Open file...', 'Open URL...', 'Open DB...', 'Generate...', 'Undo', 'Edit...', and 'Save...'. A 'Filter' section has a 'Choose' dropdown set to 'None' and 'Apply'/'Stop' buttons. The 'Current relation' section shows 'Relation: Telco_Customer_Churn' and 'Instances: 7043'. The 'Attributes' section shows 'Attributes: 21' and 'Sum of weights: 7043'. Below this is a list of attributes with checkboxes: 'customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', and 'MonthlyCharges'. A 'Remove' button is at the bottom of the list. On the right, the 'Selected attribute' section shows 'Name: customerID', 'Missing: 0 (0%)', 'Distinct: 7043', and 'Type: Nominal Unique: 7043 (100%)'. Below this is a table with 4 columns: 'No.', 'Label', 'Count', and 'Weight'. The table contains 10 rows of data. At the bottom right, there is a 'Class: Churn (Nom)' dropdown and a 'Visualize All' button. The status bar at the bottom shows 'Status OK' and a 'Log' button.

No.	Label	Count	Weight
1	7590-VHVEG	1	1
2	5575-GNVDE	1	1
3	3668-QPYBK	1	1
4	7795-CFOCW	1	1
5	9237-HQITU	1	1
6	9305-CDSKC	1	1
7	1452-KIOVK	1	1
8	6713-OKOMC	1	1
9	7892-POOKP	1	1
10	6388-TABGU	1	1



10	MonthlyCharges	0.0626442
11	TotalCharges	0.0542348
12	StreamingMovies	0.0461676
13	StreamingTV	0.0460335
14	PaperlessBilling	0.0276917
15	Dependents	0.0208718
16	Partner	0.0165241
17	SeniorCitizen	0.0152598
18	MultipleLines	0.001156
19	PhoneService	0.0001041
20	gender	0.0000535

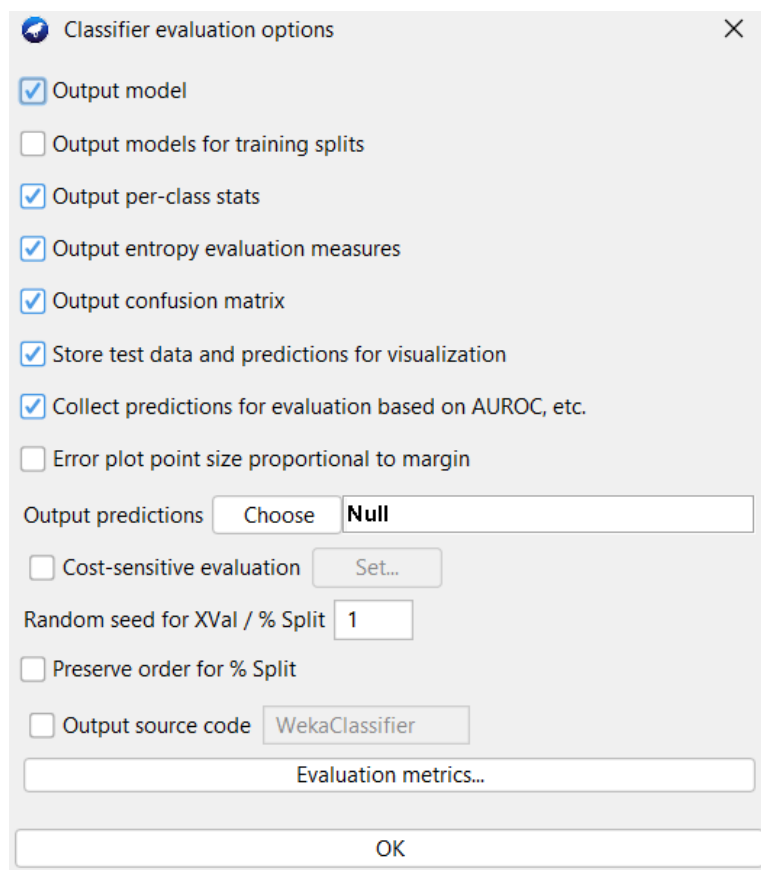
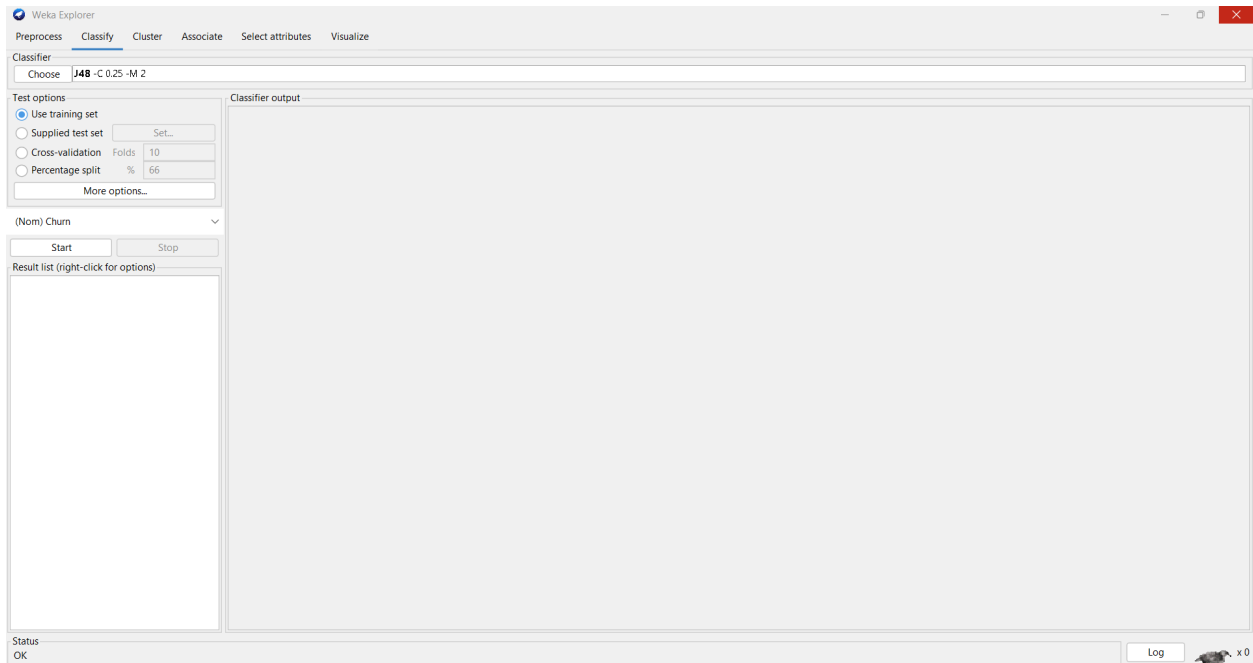
REFLECTION:

A. What could be the impact if different field were selected under the “Class name”? (5 pts)

If I selected the wrong field as the class name, the model would end up predicting something else instead of churn. This would change the whole purpose of my analysis and give results that do not match the problem I am trying to solve. Even if the system shows accuracy values, they would not be meaningful because they are based on the wrong target. I realized that this mistake can waste time and effort since the outcome would not provide any useful insights. That is why I need to make sure the correct class name is chosen before running the model.

B. Explain why it is important to select the field containing the labeled data during the Feature Selection stage. (15 pts)

I find it important to select the correct labeled field because this tells the model exactly what I want it to predict. When I choose the right label, the system can properly rank the features based on how useful they are in predicting that target. If I were to use the wrong label, the results of the feature ranking would be misleading and the model would not give reliable outcomes. I realized that this step helps me focus only on the attributes that truly matter for my analysis. By doing this, I make sure that my model becomes both accurate and meaningful.



MODELING

All

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

FilterChooseNoneApplyStop

Current relationRelation: Telco_Customer_ChurnInstances: 7043Attributes: 21Sum of weights: 7043

AttributesAllNoneInvertPattern

No.	Name
1	☒ customerID
2	☐ gender
3	☐ SeniorCitizen
4	☐ Partner
5	☐ Dependents
6	☐ tenure
7	☐ PhoneService
8	☐ MultipleLines
9	☐ InternetService
10	☐ OnlineSecurity
11	☐ OnlineBackup
12	☐ DeviceProtection
13	☐ TechSupport
14	☐ StreamingTV
15	☐ StreamingMovies
16	☐ Contract
17	☐ PaperlessBilling
18	☐ PaymentMethod
19	☐ MonthlyCharges

Remove

Selected attributeName: customerIDMissing: 0 (0%)Distinct: 7043Type: NominalUnique: 7043 (100%)

No.	Label	Count	Weight
1	7590-VHVEG	1	1
2	5575-GNVDE	1	1
3	3668-QPYBK	1	1
4	7795-CFOCW	1	1
5	9237-HQITU	1	1
6	9305-CDSKC	1	1
7	1452-KIOVK	1	1
8	6713-OKOMC	1	1
9	7892-POOKP	1	1
10	6388-TABGU	1	1

Class: Churn (Nom)Visualize All

Too many values to display.

StatusOKLog x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds
- ☐ Percentage split %

(Nom) Churn

Result list (right-click for options)

09:38:11 - trees.J48

Classifier output

Time taken to test model on training data: 0.02 seconds

=== Summary ===

Correctly Classified Instances	6078	86.2985 %
Incorrectly Classified Instances	965	13.7015 %
Kappa statistic	0.6245	
K&B Relative Info Score	45.9788 %	
K&B Information Score	2703.1356 bits	0.3838 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	3571.0944 bits	0.507 bits/instance
Complexity improvement (sf)	2307.9932 bits	0.3277 bits/instance
Mean absolute error	0.2156	
Root mean squared error	0.3283	
Relative absolute error	55.294 %	
Root relative squared error	74.3629 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.943	0.358	0.879	0.943	0.910	0.631	0.878	0.936	No
	0.642	0.057	0.803	0.642	0.713	0.631	0.878	0.766	Yes
Weighted Avg.	0.863	0.278	0.859	0.863	0.858	0.631	0.878	0.891	

=== Confusion Matrix ===

```

a      b      <-- classified as
4879  295 |   a = No
670 1199 |   b = Yes

```

Status: OK  x0

Metric	Value
Correctly Classified Instances	6078 (86.30%)
Incorrectly Classified Instances	965 (13.70%)
Kappa statistics	0.6245
Precision (Class: NO)	0.879
Precision (Class: YES)	0.803
Recall (Class: NO)	0.943
Recall (Class: YES)	0.642
F-Measures (Class: NO)	0.910
F-Measures (Class: YES)	0.713

NaiveBayes

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

10:26:56 - trees.J48

10:27:02 - bayes.NaiveBayes

Classifier output

```

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances      5281      74.9823 %
Incorrectly Classified Instances    1762      25.0177 %
Kappa statistic                    0.4624
K&B Relative Info Score            28.6919 %
K&B Information Score              1686.8198 bits    0.2395 bits/instance
Class complexity | order 0         5879.0877 bits    0.8347 bits/instance
Class complexity | scheme           9096.7074 bits    1.2916 bits/instance
Complexity improvement (SF)        -3217.6197 bits    -0.4569 bits/instance
Mean absolute error                0.2496
Root mean squared error            0.4508
Relative absolute error            64.0199 %
Root relative squared error        102.0892 %
Total Number of Instances          7043

=== Detailed Accuracy By Class ===

              TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
              0.720   0.169   0.922     0.720   0.809     0.493   0.866   0.940   No
              0.831   0.280   0.518     0.831   0.638     0.493   0.866   0.754   Yes
Weighted Avg.   0.750   0.198   0.815     0.750   0.764     0.493   0.866   0.890

=== Confusion Matrix ===

  a    b  <-- classified as
3727 1447 |    a = No
315  1554 |    b = Yes

```

Status OK

Log x 0

Metric	Value
Correctly Classified Instances	5281 (74.98%)
Incorrectly Classified Instances	1762 (25.02%)
Kappa statistic	0.462
Precision (Class: NO)	0.922
Precision (Class: YES)	0.518
Recall (Class: NO)	0.720
Recall (Class: YES)	0.831
F-Measures (Class: NO)	0.809
F-Measures (Class: YES)	0.638

DecisionTable

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **DecisionTable** -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"

Test options:

- ☒ Use training set
- ☐ Supplied test set (Set...)
- ☐ Cross-validation (Folds: 10)
- ☐ Percentage split (%: 66)
- More options...

(Nom) Churn

Start Stop

Result list (right-click for options):

- 10:26:56 - trees.J48
- 10:27:02 - bayes.NaiveBayes
- 10:29:06 - rules.DecisionTable**

Classifier output:

```
Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances      5677      80.6049 %
Incorrectly Classified Instances    1366      19.3951 %
Kappa statistic                    0.4678
K&B Relative Info Score            25.2629 %
K&B Information Score              1485.2256 bits
Class complexity | order 0         5879.0877 bits
Class complexity | scheme          4183.6217 bits
Complexity improvement (SF)        1695.4659 bits
Mean absolute error                 0.2841
Root mean squared error             0.3658
Relative absolute error             72.8588 %
Root relative squared error         82.8486 %
Total Number of Instances          7043

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
          0.905   0.468   0.843     0.905   0.873     0.473   0.853    0.940    No
          0.532   0.095   0.669     0.532   0.593     0.473   0.853    0.661    Yes
Weighted Avg.   0.806   0.369   0.797     0.806   0.798     0.473   0.853    0.866

=== Confusion Matrix ===

      a    b  <-- classified as
4682  492 |    a = No
 874   995 |    b = Yes
```

Status: OK

Log  x 0

Metric	Value
Correctly Classified Instances	5677 (80.60%)
Incorrectly Classified Instances	1366 (19.40%)
Kappa statistics	0.4678
Precision (Class: NO)	0.843
Precision (Class: YES)	0.669
Recall (Class: NO)	0.905
Recall (Class: YES)	0.532
F-Measures (Class: NO)	0.873
F-Measures (Class: YES)	0.593

LibSVM

Package Manager

Official

Refresh repository cache Install Uninstall Toggle load

☐ Installed ☒ Available ☐ All ☐ Ignore dependencies/conflicts

Install/Uninstall/Refresh progress

Package(s) installed successfully.

Unofficial File/URL

Package	Category	Installed version	Repository version	Loaded
LibLINEAR	Classification	1.9.8		
LibSVM	Classification, Regression	1.0.10	1.0.10	Yes

Package search LibSVM Clear (Search hits: 2)

LibSVM: A wrapper class for the libsvm tools

URL: <https://www.csie.ntu.edu.tw/~cjlin/libsvm/>

Author: FracPete <fracpete[at]waikato.ac.nz>

Maintainer: FracPete <fracpete[at]waikato.ac.nz>

A wrapper class for the libsvm library, including support for one-class SVMs.

All available versions:

[Latest](#)

[1.0.10](#)

[1.0.9](#)

[1.0.8](#)

[1.0.7](#)

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **LibSVM** -S 0 -K 2 -D 3 -G 0.0 -R 0.0 -N 0.5 -M 40.0 -C 1.0 -E 0.001 -P 0.1 -model "C:\Program Files\Weka-3-8-6" -seed 1

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

09:55:06 - functions.LibSVM

Classifier output

Time taken to test model on training data: 2.7 seconds

=== Summary ===

Correctly Classified Instances	5588	79.3412 %
Incorrectly Classified Instances	1455	20.6588 %
Kappa statistic	0.3905	
K&B Relative Info Score	41.6276 %	
K&B Information Score	2447.3213 bits	0.3475 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	1562670 bits	221.8756 bits/instance
Complexity improvement (SE)	-1556790.9123 bits	-221.0409 bits/instance
Mean absolute error	0.2066	
Root mean squared error	0.4545	
Relative absolute error	52.981 %	
Root relative squared error	102.942 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.933	0.594	0.813	0.933	0.869	0.412	0.670	0.808	No
	0.406	0.067	0.688	0.406	0.511	0.412	0.670	0.437	Yes
Weighted Avg.	0.793	0.454	0.780	0.793	0.774	0.412	0.670	0.709	

=== Confusion Matrix ===

```
a    b    <-- classified as
4829 345 | a = No
1110 759 | b = Yes
```

Status OK

Log x 0

Metric	Value
Correctly Classified Instances	5588 (79.34%)
Incorrectly Classified Instances	1455 (20.66%)
Kappa statistics	0.3905
Precision (Class: NO)	0.813
Precision (Class: YES)	0.688
Recall (Class: NO)	0.933
Recall (Class: YES)	0.406
F-Measures (Class: NO)	0.869
F-Measures (Class: YES)	0.511

TOP 14

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Open file...

Open URL...

Open DB...

Generate...

Undo

Edit...

Save...

Filter

Choose

None

Apply

Stop

Current relation

Relation: Telco_Customer_Churn-weka.filters.unsupervised.attrib...

Attributes: 15

Instances: 7043

Sum of weights: 7043

Attributes

All

None

Invert

Pattern

No.	Name
1	☒ customerID
2	☐ tenure
3	☐ InternetService
4	☐ OnlineSecurity
5	☐ OnlineBackup
6	☐ DeviceProtection
7	☐ TechSupport
8	☐ StreamingTV
9	☐ StreamingMovies
10	☐ Contract
11	☐ PaperlessBilling
12	☐ PaymentMethod
13	☐ MonthlyCharges
14	☐ TotalCharges
15	☐ Churn

Remove

Selected attribute

Name: customerID

Missing: 0 (0%)

Distinct: 7043

Type: Nominal

Unique: 7043 (100%)

No.	Label	Count	Weight
1	7590-VHVEG	1	1
2	5575-GNVDE	1	1
3	3668-QPYBK	1	1
4	7795-CFOCW	1	1
5	9237-HQITU	1	1
6	9305-CDSKC	1	1
7	1452-KIOVK	1	1
8	6713-OKOMC	1	1
9	7892-POOKP	1	1
10	6388-TABGU	1	1

Class: Churn (Nom)

Visualize All

Too many values to display.

Status

OK

Log

x 0

J48

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

10:08:57 - trees.J48

10:09:51 - bayes.NaiveBayes

Classifier output

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances	5999	85.1768 %
Incorrectly Classified Instances	1044	14.8232 %
Kappa statistic	0.6014	
K&B Relative Info Score	42.8807 %	
K&B Information Score	2520.9932 bits	0.3579 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	3696.1341 bits	0.5248 bits/instance
Complexity improvement (sf)	2182.9536 bits	0.3099 bits/instance
Mean absolute error	0.2254	
Root mean squared error	0.3357	
Relative absolute error	57.8012 %	
Root relative squared error	76.0301 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.925	0.352	0.879	0.925	0.902	0.605	0.876	0.936	No
	0.648	0.075	0.758	0.648	0.699	0.605	0.876	0.740	Yes
Weighted Avg.	0.852	0.278	0.847	0.852	0.848	0.605	0.876	0.884	

=== Confusion Matrix ===

```

a      b      <-- classified as
4787  387 |      a = No
657  1212 |      b = Yes

```

Status OK

Log x 0

Metric	Value
Correctly Classified Instances	5999 (85.18%)
Incorrectly Classified Instances	1044 (14.82%)
Kappa statistics	0.6014
Precision (Class: NO)	0.879
Precision (Class: YES)	0.758
Recall (Class: NO)	0.925
Recall (Class: YES)	0.648
F-Measures (Class: NO)	0.902
F-Measures (Class: YES)	0.699

NaiveBayes

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

10:08:57 - trees.J48

10:09:51 - bayes.NaiveBayes

Classifier output

=== Evaluation on training set ===

Time taken to test model on training data: 0.02 seconds

=== Summary ===

Correctly Classified Instances	5265	74.7551 %
Incorrectly Classified Instances	1778	25.2449 %
Kappa statistic	0.4609	
K&B Relative Info Score	27.8064 %	
K&B Information Score	1634.7601 bits	0.2321 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	8829.495 bits	1.2537 bits/instance
Complexity improvement (SF)	-2950.4074 bits	-0.4189 bits/instance
Mean absolute error	0.2528	
Root mean squared error	0.4522	
Relative absolute error	64.8397 %	
Root relative squared error	102.4227 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.715	0.161	0.925	0.715	0.806	0.493	0.869	0.940	No
	0.839	0.285	0.515	0.839	0.638	0.493	0.869	0.768	Yes
Weighted Avg.	0.748	0.194	0.816	0.748	0.762	0.493	0.869	0.894	

=== Confusion Matrix ===

```

a    b    <-- classified as
3697 1477 |    a = No
301  1568 |    b = Yes

```

Status: OK

Log x 0

Metric	Value
Correctly Classified Instances	5265 (74.76%)
Incorrectly Classified Instances	1778 (25.24%)
Kappa statistics	0.4609
Precision (Class: NO)	0.925
Precision (Class: YES)	0.515
Recall (Class: NO)	0.715
Recall (Class: YES)	0.839
F-Measures (Class: NO)	0.806
F-Measures (Class: YES)	0.638

Decision Table

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **DecisionTable** -X 1 -S "weka.attributeSelection.BestFirst-D 1 -N 5"

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds
- ☐ Percentage split %
-

(Nom) Churn

Result list (right-click for options)

- 10:08:57 - trees.J48
- 10:09:51 - bayes.NaiveBayes
- 10:12:25 - rules.DecisionTable**

Classifier output

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances	5677	80.6049 %
Incorrectly Classified Instances	1366	19.3951 %
Kappa statistic	0.4678	
K&B Relative Info Score	25.2629 %	
K&B Information Score	1485.2256 bits	0.2109 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	4183.6217 bits	0.594 bits/instance
Complexity improvement (Sf)	1695.4659 bits	0.2407 bits/instance
Mean absolute error	0.2841	
Root mean squared error	0.3658	
Relative absolute error	72.8588 %	
Root relative squared error	82.8486 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.905	0.468	0.843	0.905	0.873	0.473	0.853	0.940	No
	0.532	0.095	0.669	0.532	0.593	0.473	0.853	0.661	Yes
Weighted Avg.	0.806	0.369	0.797	0.806	0.798	0.473	0.853	0.866	

=== Confusion Matrix ===

a	b	<-- classified as
4682	492	a = No
874	995	b = Yes

Status: OK  x0

Metric	Value
Correctly Classified Instances	5677 (80.60%)
Incorrectly Classified Instances	1366 (19.40%)
Kappa statistics	0.4678
Precision (Class: NO)	0.843
Precision (Class: YES)	0.669
Recall (Class: NO)	0.905
Recall (Class: YES)	0.532
F-Measures (Class: NO)	0.873
F-Measures (Class: YES)	0.593

LibSVM

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **LibSVM** -S 0 -K 2 -D 3 -G 0.0 -R 0.0 -N 0.5 -M 40.0 -C 1.0 -E 0.001 -P 0.1 -model "C:\\Program Files\\Weka-3-8-6" -seed 1

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

- 10:08:57 - trees.J48
- 10:09:51 - bayes.NaiveBayes
- 10:12:25 - rules.DecisionTable
- 10:13:09 - functions.LibSVM**

Classifier output

Time taken to test model on training data: 2.09 seconds

=== Summary ===

Correctly Classified Instances	5589	79.3554 %
Incorrectly Classified Instances	1454	20.6446 %
Kappa statistic	0.3908	
K&B Relative Info Score	41.6677 %	
K&B Information Score	2449.6799 bits	0.3478 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	1561596 bits	221.7231 bits/instance
Complexity improvement (sf)	-1555716.9123 bits	-220.8884 bits/instance
Mean absolute error	0.2064	
Root mean squared error	0.4544	
Relative absolute error	52.9446 %	
Root relative squared error	102.9066 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.934	0.594	0.813	0.934	0.869	0.413	0.670	0.808	No
	0.406	0.066	0.688	0.406	0.511	0.413	0.670	0.437	Yes
Weighted Avg.	0.794	0.454	0.780	0.794	0.774	0.413	0.670	0.705	

=== Confusion Matrix ===

a	b	<-- classified as
4830	344	a = No
1110	759	b = Yes

Status OK Log x0

Metric	Value
Correctly Classified Instances	5589 (79.36%)
Incorrectly Classified Instances	1454 (20.64%)
Kappa statistics	0.3908
Precision (Class: NO)	0.813
Precision (Class: YES)	0.688
Recall (Class: NO)	0.934
Recall (Class: YES)	0.406
F-Measures (Class: NO)	0.869
F-Measures (Class: YES)	0.511

Top 10

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

Filter

ChooseNoneApplyStop

Current relation

Relation: Telco_Customer_Churn-weka.filters.unsupervised.attrib...Instances: 7043Attributes: 11Sum of weights: 7043

Attributes

AllNoneInvertPattern

No.	Name
1	☒ customerID
2	☐ tenure
3	☐ InternetService
4	☐ OnlineSecurity
5	☐ OnlineBackup
6	☐ DeviceProtection
7	☐ TechSupport
8	☐ Contract
9	☐ PaymentMethod
10	☐ MonthlyCharges
11	☐ Churn

Remove

Selected attribute

Name: customerIDMissing: 0 (0%)Distinct: 7043Type: NominalUnique: 7043 (100%)

No.	Label	Count	Weight
1	7590-VHVEG	1	1
2	5575-GNVDE	1	1
3	3668-QPYBK	1	1
4	7795-CFOCW	1	1
5	9237-HQITU	1	1
6	9305-CDSKC	1	1
7	1452-KIOVK	1	1
8	6713-OKOMC	1	1
9	7892-POOKP	1	1
10	6388-TABGU	1	1

Class: Churn (Nom)Visualize All

Too many values to display.

StatusOKLog x 0

J48

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds
- ☐ Percentage split %
-

(Nom) Churn

Result list (right-click for options)

10:18:54 - trees.J48

Classifier output

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	5820	82.6352 %
Incorrectly Classified Instances	1223	17.3648 %
Kappa statistic	0.5272	
K&B Relative Info Score	34.6714 %	
K&B Information Score	2038.3591 bits	0.2894 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	4083.0951 bits	0.5797 bits/instance
Complexity improvement (SF)	1795.9926 bits	0.255 bits/instance
Mean absolute error	0.2545	
Root mean squared error	0.3567	
Relative absolute error	65.2787 %	
Root relative squared error	80.7985 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.915	0.418	0.858	0.915	0.886	0.532	0.850	0.924	No
	0.582	0.085	0.711	0.582	0.640	0.532	0.850	0.679	Yes
Weighted Avg.	0.826	0.330	0.819	0.826	0.820	0.532	0.850	0.859	

=== Confusion Matrix ===

a	b	<-- classified as	
4732	442	a = No	
781	1088	b = Yes	

Status: OK

 x 0

Metric	Value
Correctly Classified Instances	5820 (82.64%)
Incorrectly Classified Instances	1223 (17.36%)
Kappa statistics	0.5272
Precision (Class: NO)	0.858
Precision (Class: YES)	0.711
Recall (Class: NO)	0.915
Recall (Class: YES)	0.582
F-Measures (Class: NO)	0.886
F-Measures (Class: YES)	0.640

NaiveBayes

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds: 10
- ☐ Percentage split %: 66
-

(Nom) Churn

Result list (right-click for options)

- 10:18:54 - trees.J48
- 10:19:42 - bayes.NaiveBayes

Classifier output

```
Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances      5416      76.899 %
Incorrectly Classified Instances    1627      23.101 %
Kappa statistic                    0.4952
K&B Relative Info Score            32.8226 %
K&B Information Score              1929.6719 bits
Class complexity | order 0         5879.0877 bits
Class complexity | scheme          6734.9664 bits
Complexity improvement (Sf)        -855.8787 bits
Mean absolute error                 0.2357
Root mean squared error             0.4203
Relative absolute error             60.4415 %
Root relative squared error         95.1955 %
Total Number of Instances          7043

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MDC     ROC Area  PRC Area  Class
                0.744    0.163    0.927     0.744    0.826      0.522    0.879    0.946    No
                0.837    0.256    0.542     0.837    0.658      0.522    0.879    0.783    Yes
Weighted Avg.   0.769    0.188    0.825     0.769    0.781      0.522    0.879    0.903

=== Confusion Matrix ===

  a    b  <-- classified as
3852 1322 |   a = No
305  1564 |   b = Yes
```

Status: OK x 0

Metric	Value
Correctly Classified Instances	5416 (76.90%)
Incorrectly Classified Instances	1627 (23.10%)
Kappa statistic	0.4952
Precision (Class: NO)	0.927
Precision (Class: YES)	0.542
Recall (Class: NO)	0.744
Recall (Class: YES)	0.837
F-Measures (Class: NO)	0.826

F-Measures (Class: YES)

0.658

DecisionTable

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose DecisionTable -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

- 10:18:54 - trees.J48
- 10:19:42 - bayes.NaiveBayes
- 10:21:53 - rules.DecisionTable

Classifier output

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances	5626	79.8807 %
Incorrectly Classified Instances	1417	20.1193 %
Kappa statistic	0.428	
K&B Relative Info Score	26.1694 %	
K&B Information Score	1538.5206 bits	0.2184 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	4215.7095 bits	0.5986 bits/instance
Complexity improvement (Sf)	1663.3782 bits	0.2362 bits/instance
Mean absolute error	0.2804	
Root mean squared error	0.3683	
Relative absolute error	71.9138 %	
Root relative squared error	83.4209 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.918	0.532	0.827	0.918	0.870	0.440	0.846	0.937	No
	0.468	0.082	0.674	0.468	0.552	0.440	0.846	0.641	Yes
Weighted Avg.	0.799	0.413	0.786	0.799	0.786	0.440	0.846	0.858	

=== Confusion Matrix ===

a	b	<-- classified as
4752	422	a = No
995	874	b = Yes

Status OK

Log x 0

Metric	Value
Correctly Classified Instances	5626 (79.88%)
Incorrectly Classified Instances	1417 (20.12%)
Kappa statistic	0.428
Precision (Class: NO)	0.827
Precision (Class: YES)	0.674
Recall (Class: NO)	0.918
Recall (Class: YES)	0.468

F-Measures (Class: NO)	0.870
F-Measures (Class: YES)	0.552

LibSVM

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **LibSVM** -S 0 -K 2 -D 3 -G 0.0 -R 0.0 -N 0.5 -M 40.0 -C 1.0 -E 0.001 -P 0.1 -model "C:\\Program Files\\Weka-3-8-6" -seed 1

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

- 10:18:54 - trees.J48
- 10:19:42 - bayes.NaiveBayes
- 10:21:53 - rules.DecisionTable
- 10:23:04 - functions.LibSVM

Classifier output

Time taken to test model on training data: 1.38 seconds

=== Summary ===

Correctly Classified Instances	5574	79.1424 %
Incorrectly Classified Instances	1469	20.8576 %
Kappa statistic	0.3836	
K&B Relative Info Score	41.0659 %	
K&B Information Score	2414.3009 bits	0.3428 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	1577706 bits	224.0105 bits/instance
Complexity improvement (sf)	-1571826.9123 bits	-223.1758 bits/instance
Mean absolute error	0.2086	
Root mean squared error	0.4567	
Relative absolute error	53.4908 %	
Root relative squared error	103.436 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.933	0.600	0.812	0.933	0.868	0.406	0.666	0.806	No
	0.400	0.067	0.682	0.400	0.505	0.406	0.666	0.432	Yes
Weighted Avg.	0.791	0.458	0.777	0.791	0.771	0.406	0.666	0.707	

=== Confusion Matrix ===

a	b	<-- classified as
4926	348	a = No
1121	748	b = Yes

Status OK

Log x 0

Metric	Value
Correctly Classified Instances	5574 (79.14%)
Incorrectly Classified Instances	1469 (20.86%)
Kappa statistic	0.384
Precision (Class: NO)	0.812
Precision (Class: YES)	0.682
Recall (Class: NO)	0.933

Recall (Class: YES)	0.400
F-Measures (Class: NO)	0.868
F-Measures (Class: YES)	0.505

Top7

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

FilterChooseNoneApplyStop

Current relation
Relation: Telco_Customer_Churn-weka.filters.unsupervised.attrib...
Instances: 7043
Attributes: 8
Sum of weights: 7043

Attributes
AllNoneInvertPattern

No.	Name
1	☒ customerID
2	☐ tenure
3	☐ InternetService
4	☐ OnlineSecurity
5	☐ OnlineBackup
6	☐ TechSupport
7	☐ Contract
8	☐ Churn

Remove

Selected attribute
Name: customerID
Missing: 0 (0%)
Distinct: 7043
Type: Nominal
Unique: 7043 (100%)

No.	Label	Count	Weight
1	7590-VHVEG	1	1
2	5575-GNVDE	1	1
3	3668-QPYBK	1	1
4	7795-CFOCW	1	1
5	9237-HQITU	1	1
6	9305-CDSKC	1	1
7	1452-KIOVK	1	1
8	6713-OKOMC	1	1
9	7892-POOKP	1	1
10	6388-TABGU	1	1

Class: Churn (Nom)Visualize All

Too many values to display.

StatusOK

Log x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds
- ☐ Percentage split %

(Nom) Churn

Result list (right-click for options)

10:31:38 - trees.J48

Classifier output

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	5637	80.0369 %
Incorrectly Classified Instances	1406	19.9631 %
Kappa statistic	0.4253	
K&B Relative Info Score	23.8998 %	
K&B Information Score	1405.0898 bits	0.1995 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	4672.6305 bits	0.6634 bits/instance
Complexity improvement (sf)	1206.4571 bits	0.1713 bits/instance
Mean absolute error	0.2945	
Root mean squared error	0.3837	
Relative absolute error	75.5235 %	
Root relative squared error	86.9077 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.926	0.547	0.824	0.926	0.872	0.441	0.785	0.877	No
	0.453	0.074	0.688	0.453	0.546	0.441	0.785	0.565	Yes
Weighted Avg.	0.800	0.421	0.788	0.800	0.786	0.441	0.785	0.795	

=== Confusion Matrix ===

a	b	<-- classified as
4790	384	a = No
1022	847	b = Yes

Status: OK x 0

Metric	Value
Correctly Classified Instances	5637 (80.04%)
Incorrectly Classified Instances	1406 (19.96%)
Kappa statistic	0.425
Precision (Class: NO)	0.824
Precision (Class: YES)	0.688
Recall (Class: NO)	0.926
Recall (Class: YES)	0.453
F-Measures (Class: NO)	0.872
Correctly Classified Instances	5637 (80.04%)

NaiveBayes

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds
- ☐ Percentage split %

(Nom) Churn

Result list (right-click for options)

- 10:31:38 - trees.J48
- 10:35:17 - bayes.NaiveBayes**

Classifier output

```
Time taken to test model on training data: 0.01 seconds

=== Summary ===
Correctly Classified Instances      5495      78.0207 %
Incorrectly Classified Instances    1548      21.9793 %
Kappa statistic                    0.5119
K&B Relative Info Score            36.1718 %
K&B Information Score              2126.5697 bits
Class complexity | order 0         5879.0877 bits
Class complexity | scheme          5151.0305 bits
Complexity improvement (sf)         728.0572 bits
Mean absolute error                 0.2248
Root mean squared error             0.3965
Relative absolute error              57.648 %
Root relative squared error         89.8045 %
Total Number of Instances          7043

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      0.763    0.172    0.925     0.763    0.836      0.534    0.897    0.955    No
      0.828    0.237    0.558     0.828    0.667      0.534    0.897    0.828    Yes
Weighted Avg.   0.780    0.189    0.827     0.780    0.791      0.534    0.897    0.922

=== Confusion Matrix ===

      a    b  <-- classified as
3947 1227 |    a = No
 321 1548 |    b = Yes
```

Status: OK  x 0

Metric	Value
Correctly Classified Instances	5495 (78.02%)
Incorrectly Classified Instances	1548 (21.98%)
Kappa statistic	0.512
Precision (Class: NO)	0.925
Precision (Class: YES)	0.558
Recall (Class: NO)	0.763
Recall (Class: YES)	0.828
F-Measures (Class: NO)	0.836
F-Measures (Class: YES)	0.667

DecisionTable

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: **DecisionTable** -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"

Test options:

- ☒ Use training set
- ☐ Supplied test set (Set...)
- ☐ Cross-validation (Folds: 10)
- ☐ Percentage split (%: 66)
- More options...

(Nom) Churn

Start Stop

Result list (right-click for options):

- 10:31:38 - trees.J48
- 10:35:17 - bayes.NaiveBayes
- 10:36:33 - rules.DecisionTable**

Classifier output:

Time taken to test model on training data: 0 seconds

=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	5643	80.1221 %
Incorrectly Classified Instances	1400	19.8779 %
Kappa statistic	0.4402	
K&B Relative Info Score	26.4374	%
K&B Information Score	1554.2795	bits
Class complexity order 0	5879.0877	bits
Class complexity scheme	4247.826	bits
Complexity improvement (sf)	1631.2617	bits
Mean absolute error	0.2785	
Root mean squared error	0.3688	
Relative absolute error	71.4117	%
Root relative squared error	83.5303	%
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.915	0.514	0.831	0.915	0.871	0.450	0.842	0.933	No
	0.486	0.085	0.674	0.486	0.565	0.450	0.842	0.643	Yes
Weighted Avg.	0.801	0.400	0.790	0.801	0.790	0.450	0.842	0.856	

=== Confusion Matrix ===

a	b	<-- classified as
4735	439	a = No
961	908	b = Yes

Status: OK

Log x0

Metric	Value
Correctly Classified Instances	5643 (80.12%)
Incorrectly Classified Instances	1400 (19.88%)
Kappa statistic	0.440
Precision (Class: NO)	0.831
Precision (Class: YES)	0.674
Recall (Class: NO)	0.915
Recall (Class: YES)	0.486
F-Measures (Class: NO)	0.871
F-Measures (Class: YES)	0.565

LibSVM

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier Choose **LibSVM** -S 0 -K 2 -D 3 -G 0.0 -R 0.0 -N 0.5 -M 40.0 -C 1.0 -E 0.001 -P 0.1 -model "C:\\Program Files\\Weka-3-8-6" -seed 1

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Churn

Start Stop

Result list (right-click for options)

- 10:31:38 - trees.J48
- 10:35:17 - bayes.NaiveBayes
- 10:36:33 - rules.DecisionTable
- 10:37:03 - functions.LibSVM**

Classifier output

Time taken to test model on training data: 1.1 seconds

=== Summary ===

Correctly Classified Instances	5174	73.463 %
Incorrectly Classified Instances	1869	26.537 %
Kappa statistic	0	
K&B Relative Info Score	25.0185 %	
K&B Information Score	1470.86 bits	0.2088 bits/instance
Class complexity order 0	5879.0877 bits	0.8347 bits/instance
Class complexity scheme	2007306 bits	285.0072 bits/instance
Complexity improvement (SF)	-2001426.9123 bits	-284.1725 bits/instance
Mean absolute error	0.2654	
Root mean squared error	0.5151	
Relative absolute error	68.056 %	
Root relative squared error	116.6717 %	
Total Number of Instances	7043	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	1.000	0.735	1.000	0.847	?	0.500	0.735	No
	0.000	0.000	?	0.000	?	?	0.500	0.265	Yes
Weighted Avg.	0.735	0.735	?	0.735	?	?	0.500	0.610	

=== Confusion Matrix ===

a	b	<-- classified as
5174	0	a = No
1869	0	b = Yes

Status OK

Log x 0

Metric	Value
Correctly Classified Instances	5174 (73.46%)
Incorrectly Classified Instances	1869 (26.54%)
Kappa statistic	0.000
Precision (Class: NO)	0.735
Precision (Class: YES)	?
Recall (Class: NO)	1.000
Recall (Class: YES)	0.000
F-Measures (Class: NO)	0.847
F-Measures (Class: YES)	?

CLASSIFI ERS	Trees » J48				Bayes » NaiveBayes				Rules » DecisionTable				Functions » LibSVM			
ATTRIBU TES	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7
Correctly Classified Instances	6078	5999	5820	5637	5281	5265	5416	5495	5677	5677	5626	5643	5588	5589	5574	5174
Incorrectl y Classified Instances	965	1044	1223	1406	1762	1778	1627	1548	1366	1366	1417	1400	1455	1454	1469	1869
Kappa statistics	0.6245	0.6014	0.5272	0.425	0.462	0.4609	0.4952	0.512	0.4678	0.4678	0.428	0.44	0.3905	0.3908	0.384	0
Precision (Class: NO)	0.879	0.879	0.858	0.824	0.922	0.925	0.927	0.925	0.843	0.843	0.827	0.831	0.813	0.813	0.812	0.735
Precision (Class: YES)	0.803	0.758	0.711	0.688	0.518	0.515	0.542	0.558	0.669	0.669	0.674	0.674	0.688	0.688	0.682	?
Recall (Class: NO)	0.943	0.925	0.915	0.926	0.72	0.715	0.744	0.763	0.905	0.905	0.918	0.915	0.933	0.934	0.933	1
Recall (Class: YES)	0.642	0.648	0.582	0.453	0.831	0.839	0.837	0.828	0.532	0.532	0.468	0.486	0.406	0.406	0.4	0
F-Measur es (Class: NO)	0.91	0.902	0.886	0.872	0.809	0.806	0.826	0.836	0.873	0.873	0.87	0.871	0.869	0.869	0.868	0.847
F-Measur es (Class: YES)	0.713	0.699	0.64	0.546	0.638	0.638	0.658	0.667	0.593	0.593	0.552	0.565	0.511	0.511	0.505	?

REFLECTION:

A. Based on the assessment provided in ANNEX-01, is there any relevance in performing Feature selection prior to the Modeling phase? Elaborate your answer. (10 pts)

Yes, I believe there is a clear relevance in performing feature selection before modeling. When I applied feature selection, I noticed that the models became more focused and efficient since only the most important attributes were kept. This reduced unnecessary noise from the dataset and helped improve the overall reliability of the results. It also made the modeling process easier to manage because fewer attributes meant less complexity. From this, I realized that feature selection is an important step to make the model more accurate and meaningful.